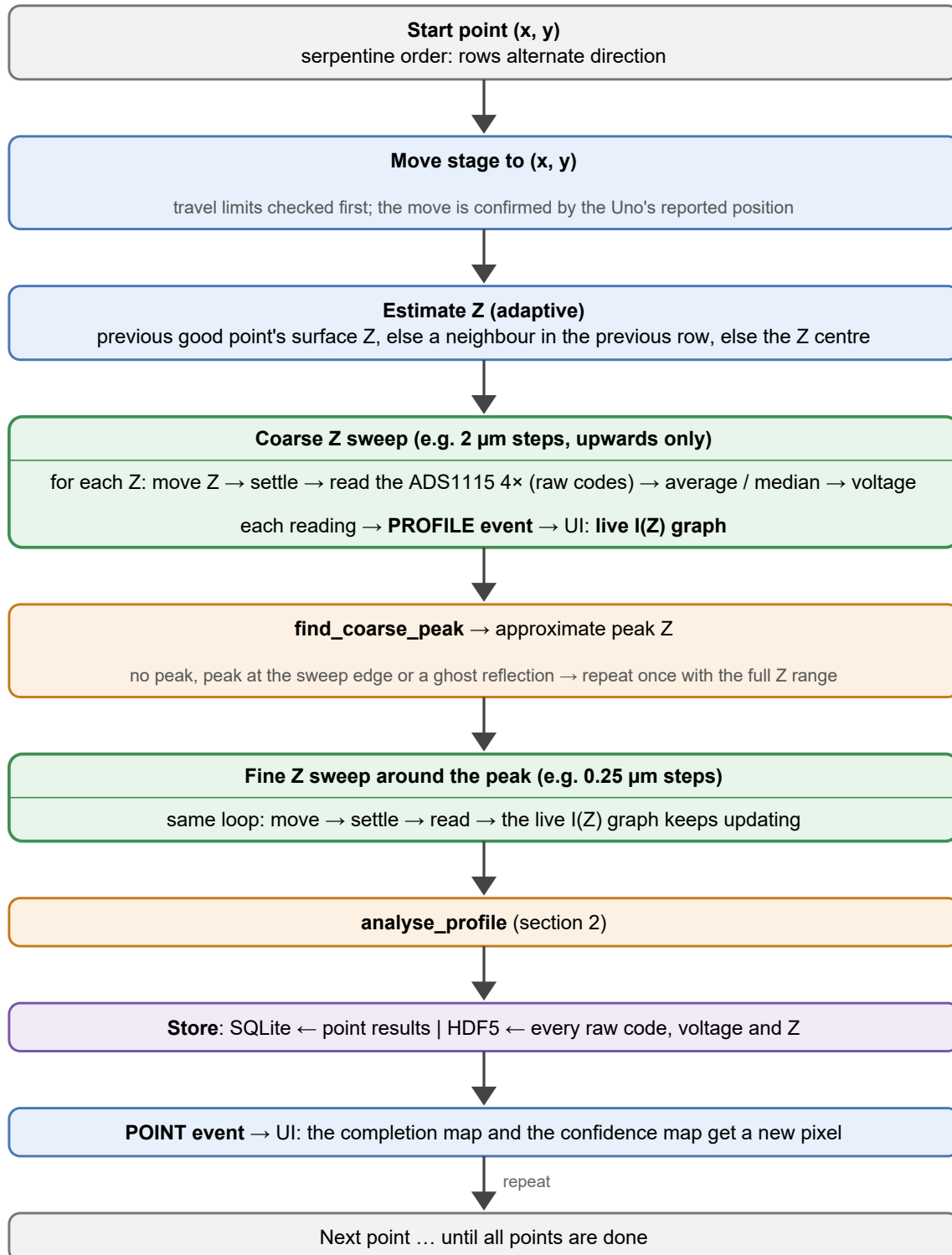


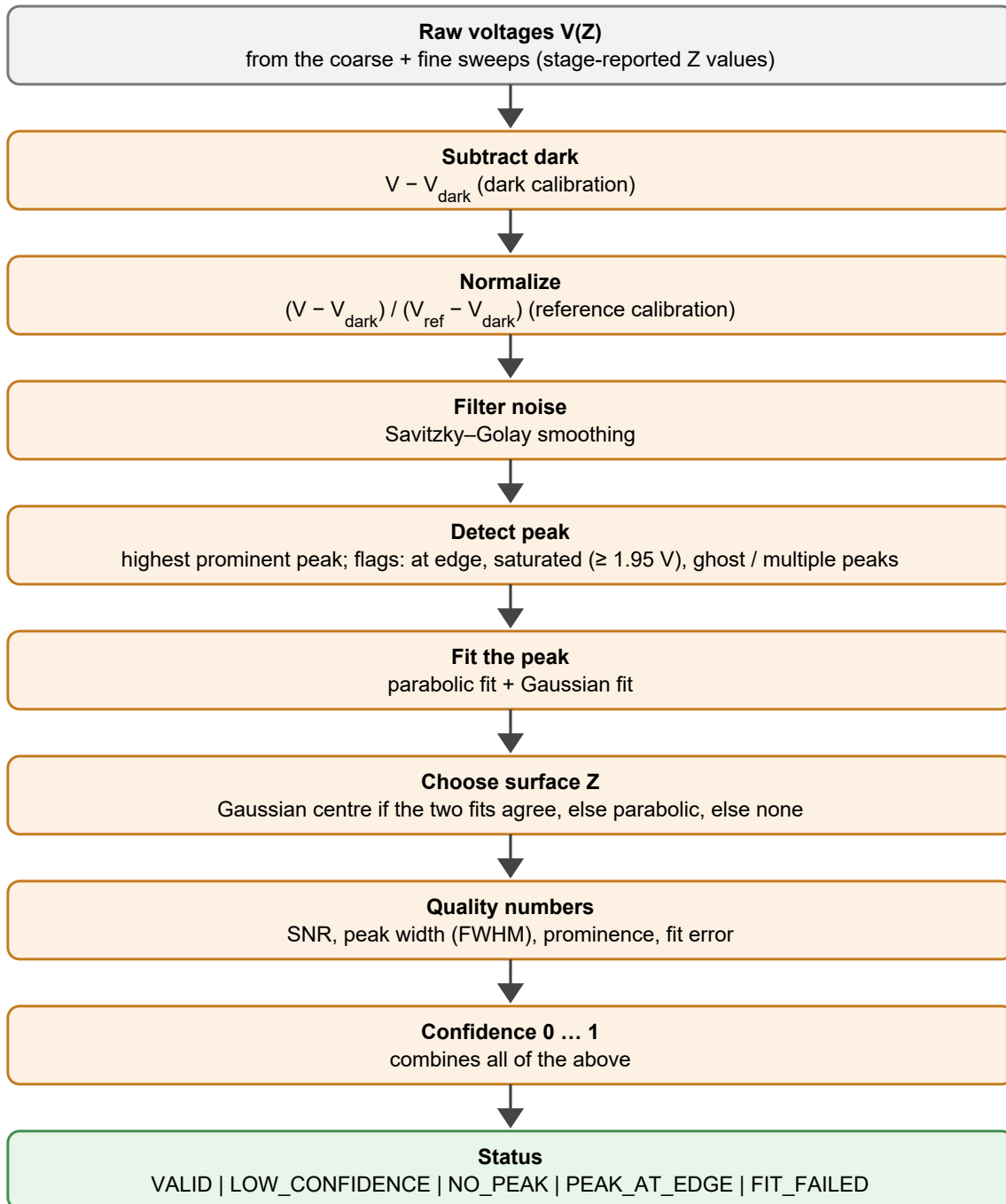
Confocal Surface Scanner: from intensity readings to a 3D surface

Data flow of the software on the Raspberry Pi 5 (OpenFlexure Delta Stage, Arduino Uno, ADS1115 + OPT101)

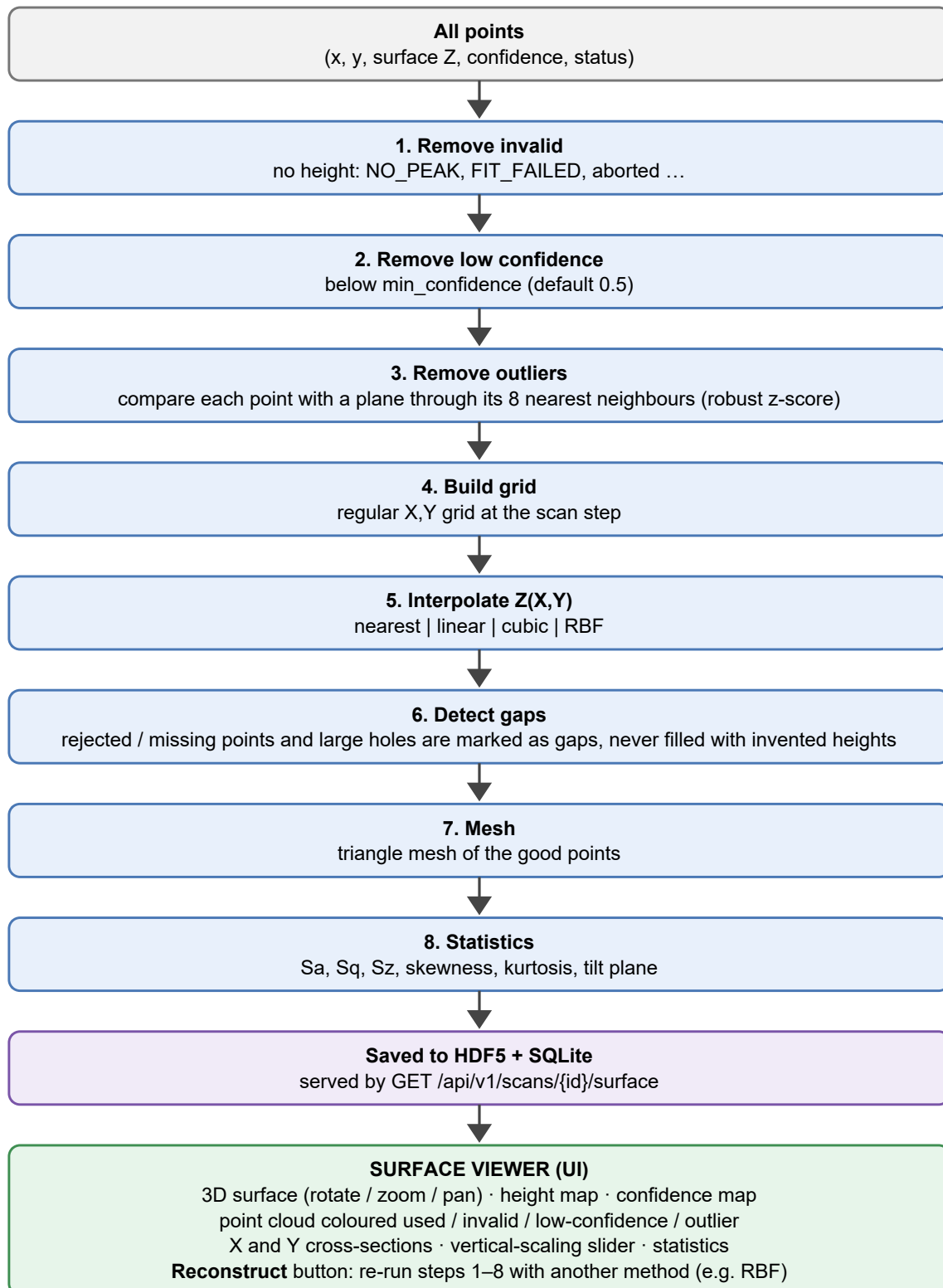
1. For every X,Y point (scan engine on the Pi)



2. Inside analyse_profile: one I(Z) curve becomes one height



3. After the last point: building the 3D surface



4. Where each graph in the UI gets its data

UI graph	Source
Live I(Z) graph (Live Scan)	PROFILE events over the WebSocket, during each sweep
Completion map + confidence map	POINT events (one per finished point), plus stored points on reload
Per-point I(Z) (Scan History)	GET .../profile/{point_id}: raw samples, averaged, normalized, filtered
3D surface, height map, cross-sections	GET .../surface: the result of section 3

Raw data is never thrown away: every ADC code at every Z is kept in the scan's HDF5 file, so the reconstruction can be re-run later with different settings without re-scanning.